

## FENTRESS NALF, VA, USA

WMO: 725172

Lat: 36.695N

Lon: 76.136W

Elev: 5

StdP: 101.27

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 13762

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -5.8       | -3.6 | -17.3                         | 0.8 | -2.5 | -14.9 | 1.0 | -1.8 | 9.1                   | 9.4  | 8.3 | 8.2  | 2.4                   | 340  | 0.401 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.2                   | 35.8            | 25.2 | 34.5 | 24.8 | 33.3 | 24.2 | 26.7                | 33.3 | 26.0 | 32.1 | 25.5 | 31.1 | 2.8                  | 200  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 24.9                            | 20.0 | 30.0 | 24.3 | 19.3 | 29.3 | 23.8 | 18.7 | 28.9 | 83.1          | 33.1 | 80.3 | 32.2 | 77.8 | 31.1 | 29.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| (4) 8.3           | 7.4  | 6.3 | DB -9.9                    | 37.8 | 4.0                | 0.9 | -12.8                                              | 38.5 | -15.1      | 39.0 | -17.4      | 39.5 | -20.2      | 40.1 |  |  |
| (5)               |      |     | WB -11.5                   | 27.6 | 3.2                | 1.3 | -13.8                                              | 28.5 | -15.6      | 29.3 | -17.4      | 30.0 | -19.7      | 30.9 |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 17.3   | 5.9   | 8.9   | 11.0  | 17.6  | 21.9  | 25.8  | 27.4  | 26.4  | 24.0  | 18.5  | 11.7  | 8.7  |
|                                                                                  | DBStd       | 8.77   | 5.66  | 6.07  | 5.34  | 5.61  | 4.21  | 3.10  | 2.69  | 2.24  | 3.10  | 4.69  | 4.71  | 5.62 |
|                                                                                  | HDD10.0     | 421    | 152   | 86    | 53    | 3     | 0     | 0     | 0     | 0     | 0     | 2     | 33    | 92   |
|                                                                                  | HDD18.3     | 1556   | 386   | 269   | 237   | 79    | 17    | 1     | 0     | 0     | 1     | 58    | 207   | 302  |
|                                                                                  | CDD10.0     | 3103   | 24    | 55    | 83    | 231   | 368   | 473   | 539   | 509   | 421   | 264   | 83    | 53   |
|                                                                                  | CDD18.3     | 1197   | 1     | 6     | 8     | 57    | 127   | 224   | 281   | 251   | 171   | 62    | 7     | 4    |
|                                                                                  | CDH23.3     | 11755  | 1     | 66    | 84    | 666   | 1136  | 2309  | 3138  | 2427  | 1421  | 453   | 31    | 24   |
|                                                                                  | CDH26.7     | 4841   | 0     | 16    | 21    | 306   | 437   | 1004  | 1459  | 984   | 486   | 124   | 3     | 2    |
| Wind                                                                             | WSAvg       | 2.5    | 2.6   | 2.9   | 3.2   | 3.2   | 2.6   | 2.2   | 2.0   | 1.8   | 2.3   | 2.3   | 2.4   | 2.3  |
| Precipitation                                                                    | PrecAvg     | 1231   | 85    | 77    | 93    | 85    | 91    | 112   | 144   | 138   | 136   | 102   | 81    | 87   |
|                                                                                  | PrecMax     | 1589   | 160   | 199   | 234   | 164   | 137   | 288   | 243   | 325   | 385   | 212   | 271   | 188  |
|                                                                                  | PrecMin     | 779    | 27    | 22    | 19    | 19    | 36    | 42    | 38    | 43    | 14    | 3     | 4     | 37   |
|                                                                                  | PrecStd     | 199    | 34    | 39    | 41    | 35    | 30    | 51    | 50    | 69    | 92    | 54    | 58    | 39   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 23.1  | 29.3  | 28.9  | 38.8  | 35.6  | 36.7  | 37.2  | 35.5  | 34.4  | 32.6  | 26.5  | 26.2 |
|                                                                                  |             | MCWB   | 17.7  | 16.9  | 17.9  | 23.4  | 23.3  | 25.5  | 26.7  | 25.9  | 25.2  | 23.1  | 20.2  | 21.0 |
|                                                                                  | 2%          | DB     | 20.4  | 25.1  | 25.6  | 33.2  | 32.8  | 34.9  | 35.8  | 34.1  | 32.8  | 29.7  | 23.8  | 23.1 |
|                                                                                  |             | MCWB   | 15.4  | 16.9  | 16.0  | 21.3  | 22.2  | 24.5  | 26.2  | 25.3  | 24.2  | 21.9  | 18.3  | 18.8 |
|                                                                                  | 5%          | DB     | 18.0  | 22.3  | 22.9  | 30.0  | 31.1  | 33.5  | 34.5  | 32.9  | 31.3  | 27.8  | 21.8  | 20.6 |
|                                                                                  |             | MCWB   | 13.4  | 15.1  | 15.2  | 19.4  | 22.1  | 24.1  | 25.4  | 24.6  | 23.5  | 21.2  | 17.2  | 17.2 |
|                                                                                  | 10%         | DB     | 15.2  | 19.6  | 20.6  | 27.0  | 29.2  | 31.9  | 33.0  | 31.5  | 29.6  | 25.8  | 19.7  | 18.2 |
|                                                                                  |             | MCWB   | 11.3  | 14.6  | 14.7  | 18.2  | 21.0  | 23.4  | 24.8  | 24.0  | 22.8  | 20.3  | 15.2  | 15.3 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 19.5  | 19.9  | 19.6  | 24.1  | 24.6  | 26.5  | 27.9  | 27.1  | 26.4  | 25.2  | 21.7  | 21.6 |
|                                                                                  |             | MCDB   | 22.2  | 25.0  | 25.9  | 33.9  | 30.5  | 33.2  | 35.5  | 33.3  | 31.5  | 29.8  | 24.5  | 25.0 |
|                                                                                  | 2%          | WB     | 16.5  | 18.5  | 18.2  | 22.6  | 23.8  | 25.7  | 27.0  | 26.3  | 25.6  | 23.9  | 19.7  | 19.4 |
|                                                                                  |             | MCDB   | 19.7  | 23.0  | 23.5  | 32.0  | 29.6  | 32.3  | 34.2  | 32.4  | 30.7  | 27.8  | 23.0  | 22.4 |
|                                                                                  | 5%          | WB     | 14.2  | 17.0  | 16.7  | 20.9  | 23.0  | 25.1  | 26.3  | 25.7  | 24.9  | 22.5  | 18.0  | 17.7 |
|                                                                                  |             | MCDB   | 17.3  | 21.7  | 21.9  | 26.9  | 29.0  | 31.7  | 32.9  | 30.9  | 29.5  | 26.3  | 21.0  | 20.4 |
|                                                                                  | 10%         | WB     | 11.4  | 14.8  | 14.7  | 19.3  | 22.3  | 24.5  | 25.6  | 25.0  | 24.1  | 21.2  | 15.9  | 15.5 |
|                                                                                  |             | MCDB   | 14.9  | 19.1  | 19.7  | 25.5  | 27.7  | 30.3  | 31.5  | 29.6  | 27.8  | 24.9  | 19.4  | 18.4 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 10.7  | 11.7  | 11.5  | 12.2  | 10.8  | 10.5  | 10.2  | 9.7   | 9.6   | 11.1  | 11.0  | 10.1 |
|                                                                                  |             | MCDBR  | 14.7  | 15.9  | 15.4  | 15.7  | 12.6  | 11.5  | 11.2  | 11.0  | 11.4  | 11.9  | 12.2  | 10.9 |
|                                                                                  |             | MCWBR  | 11.0  | 9.3   | 8.6   | 6.8   | 5.2   | 4.3   | 4.1   | 4.2   | 4.9   | 5.7   | 8.1   | 8.5  |
|                                                                                  | 5% WB       | MCDBR  | 12.6  | 14.7  | 13.2  | 12.3  | 10.7  | 10.4  | 10.3  | 9.9   | 9.1   | 9.3   | 10.3  | 10.5 |
| MCWBR                                                                            |             | 11.1   | 10.0  | 9.1   | 6.8   | 4.8   | 4.3   | 4.0   | 4.1   | 4.3   | 4.9   | 8.1   | 8.8   |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.311  | 0.323 | 0.349 | 0.391 | 0.435 | 0.484 | 0.509 | 0.502 | 0.407 | 0.358 | 0.327 | 0.316 |      |
|                                                                                  | taud        | 2.536  | 2.497 | 2.453 | 2.349 | 2.256 | 2.173 | 2.114 | 2.122 | 2.382 | 2.508 | 2.544 | 2.553 |      |
|                                                                                  | Ebn at Noon | 887    | 916   | 918   | 891   | 852   | 808   | 784   | 780   | 846   | 864   | 860   | 857   |      |
|                                                                                  | Edh at Noon | 78     | 91    | 105   | 123   | 137   | 149   | 157   | 152   | 111   | 90    | 77    | 73    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.35   | 3.11  | 4.17  | 5.32  | 5.99  | 6.39  | 6.09  | 5.49  | 4.49  | 3.57  | 2.66  | 2.10  |      |
|                                                                                  | RadStd      | 0.19   | 0.33  | 0.32  | 0.38  | 0.49  | 0.40  | 0.30  | 0.33  | 0.34  | 0.40  | 0.21  | 0.17  |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (51 neighbors) |  | +0.45 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +168    | +108    |

Nomenclature: See separate page